

Multiclass Classification of Fetal Health Using Cardiotocogram Data

Course: UE24CS352A — Machine Learning
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1. Problem Statement & Clinical Motivation

Fetal well-being assessment is among the most sensitive diagnostic challenges in obstetrics. Intrapartum complications remain a leading contributor to preventable neonatal morbidity and mortality worldwide. Cardiotocography (CTG)—which simultaneously monitors fetal heart rate (FHR) dynamics and uterine contractions (UC)—is the standard clinical modality for identifying hypoxemia, acidosis, and fetal distress. However, traditional visual CTG interpretation by clinicians suffers from substantial intra- and inter-observer variability, creating diagnostic bottlenecks and delays in acute medical interventions. This project develops an end-to-end, high-precision machine learning pipeline to automatically categorize fetal health status from multidimensional CTG recordings into three clinically recognized states: Normal, Suspect, or Pathological.

2. Dataset Description & Preprocessing Pipeline

The study employs the benchmark open-access Cardiotocography dataset compiled by Ayres-de-Campos et al. via the UCI ML Repository and Kaggle. It comprises 2,126 fetal examination recordings described across 21 continuous morphological and statistical features.

- **Feature Topology:** Encompasses baseline FHR (bpm), accelerations, fetal movements, uterine contractions, light decelerations, severe decelerations, prolonged decelerations, short-term and long-term FHR variability indices, and 10 histogram-derived morphological metrics (width, min, max, peaks, mode, mean, median, variance, and tendency).
- **Target Distribution ('fetal_health'):** Trimodal classification: Class 1 (Normal) = 1,646 instances (77.9%), Class 2 (Suspect) = 292 instances (13.8%), and Class 3 (Pathological) = 175 instances (8.3%). This reflects real-world clinical class imbalance.
- **Data Cleansing & Stratification:** Initial inspection confirmed zero missing/null values across all columns. A total of 13 exact duplicate records were pruned, yielding 2,113 unique records. Features were standardized via StandardScaler to zero mean and unit variance. A stratified 70/30 train-test partition was executed (1,479 train / 634 test), preserving exact class ratios across subsets.

3. Exploratory Data Analysis & Feature Importance

Exploratory Data Analysis (EDA) revealed vital physiological insights. Feature correlation analysis using Pearson heatmaps demonstrated strong collinearity among histogram centrality metrics (mean, median, mode) and inverse relationships between FHR variability and pathological distress. Feature importance ranking using ANOVA F-value SelectKBest established that 'prolongued_decelerations' (F-score: 505.85), 'percentage_of_time_with_abnormal_long_term_variability' (F-score: 345.16), and 'abnormal_short_term_variability' (F-score: 343.82) are the primary physiological biomarkers separating healthy fetuses from distressed or suspect cases.

4. Implementation Overview & Comparative Model Evaluation

The engineering workload was bifurcated equally between the two team members, implementing 10 distinct classification architectures ranging from linear baselines to state-of-the-art gradient-boosted ensembles.

Model Architecture	Implemented By	Category	Accuracy	Performance Notes
LightGBM	Shreya Hegde	Gradient Boost	95.89%	Best overall; leaf-wise growth captures fine non-linear thresholds
XGBoost	Naveen Prasad M	Gradient Boost	95.27%	Strong regularization; 1.00 precision on Pathological cases
Gradient Boosting	Shreya Hegde	Sequential Ensemble	95.11%	Iterative residual correction; stable and balanced recall
Random Forest	Naveen Prasad M	Bagging Ensemble	94.79%	Variance reduction via decorrelated trees; 0.97 F1 on Normal
Decision Tree (CART)	Shreya Hegde	Tree-based	92.58%	High interpretability; slightly prone to variance on minor splits
Multi-layer Perceptron	Naveen Prasad M	Neural Network	92.27%	Feedforward ANN with Adam optimizer; captures non-linear manifolds
SVM (RBF Kernel)	Naveen Prasad M	Kernel Method	91.01%	Infinite-dimensional kernel mapping; robust margin separation
Linear SVM	Shreya Hegde	Linear Max-Margin	89.90%	Linear hyperplanes struggle with overlapping boundary clusters
Logistic Regression	Naveen Prasad M	Probabilistic Linear	89.43%	Baseline generalized linear model; constrained by linearity
K-Nearest Neighbors	Shreya Hegde	Instance-based	88.80%	Euclidean metric sensitive to high-dimensional density shifts

Hyperparameter Tuning Analysis

Extensive 3-fold cross-validation grid search was conducted on the top-ranking LightGBM classifier over parameters including n_estimators (100–300), learning_rate (0.05–0.2), and num_leaves (15–63). While cross-validation attained an optimal CV accuracy of 94.25%, the tuned model recorded 95.58% on the held-out test set compared to 95.89% default. This corroborates empirical findings in medical ML literature that gradient boosted models are susceptible to in-sample overfitting during aggressive hyperparameter tuning on modest sample sizes.

5. Conclusions & Future Directions

This project successfully designed, implemented, and validated an exhaustive benchmarking pipeline for CTG fetal health classification. The empirical findings deliver concrete clinical and engineering takeaways:

- Tree Ensembles Dominate: Gradient boosting architectures (LightGBM at 95.89%, XGBoost at 95.27%, and Gradient Boosting at 95.11%) consistently outperformed linear and distance-based baselines by over 6%, effectively modeling complex non-linear clinical thresholds.
- Clinical Sensitivity: Despite severe class imbalance (8.3% Pathological prevalence), XGBoost achieved a perfect 1.00 precision and 0.96 F1-score for the Pathological class, ensuring that life-threatening fetal distress is reliably identified with near-zero false alarms.
- Future Deployment: Future extensions will incorporate synthetic minority oversampling (SMOTE) and cost-sensitive loss matrices to penalize false negatives on suspect cases, paving the way for trustworthy, real-time clinical decision support.